

USER MANUAL

Digital Communications Visualization Console

Source Coding • Channel Coding • Modulation • Noisy-Channel Simulation



1. Overview

This console is a hands-on teaching device for Telecommunications. It walks you through the full journey a message takes across a real digital link: turning letters into compact codes, packaging those codes for transmission, protecting them with error-checking, sending them through a simulated noisy channel, and decoding what arrives on the other side.

Two full-color displays, a set of navigation buttons, a numeric/letter keypad, a noise-level dial, and two rows of LED indicators let you configure each stage of the process and watch it happen step by step.

What you can do with it

- Type a short message using letters A through E.
- Choose how that message is compressed (source coding): Fixed Length, Huffman, or Shannon–Fano.
- Choose how it is protected against errors (channel coding): parity checks or CRC checksums.
- Pick a modulation style to visualize (BPSK, QPSK, ASK, PSK, or FSK).
- Turn simulated transmission acknowledgements (ACK) on or off.
- Run a simulated transmission through a noisy channel and watch it succeed, fail, or retry.
- Review exactly what was sent, what was received, and how it was decoded.

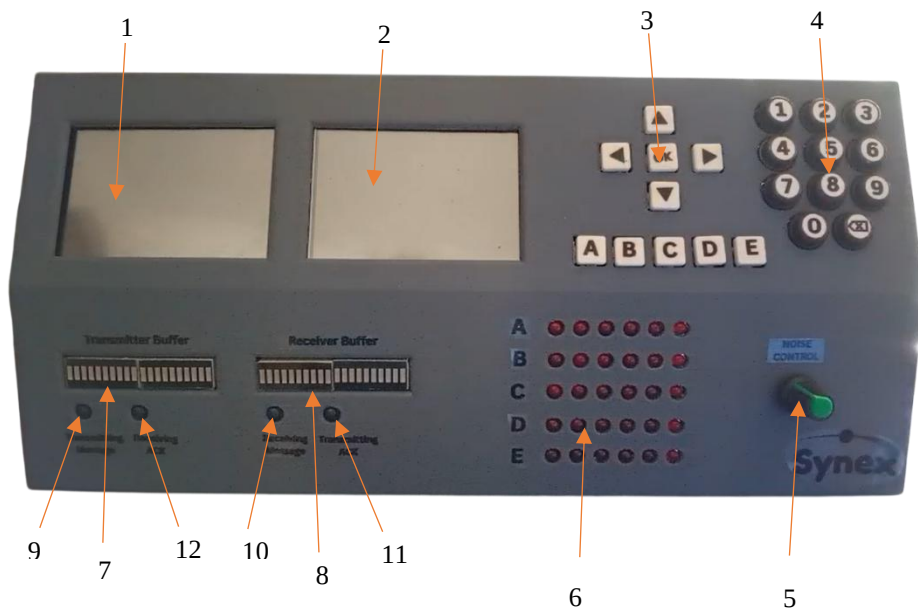
2. Safety Precautions

Please read these guidelines carefully before operating the console. Following these instructions will protect both the user and the device.

- Only use the included 12V DC power adapter. Using third-party chargers may deliver incorrect voltage and permanently damage the internal circuit boards.
- Do not block the cooling vents on the casing. The internal power regulators require continuous airflow to manage heat efficiently during long laboratory sessions.
- Keep water, coffee, and other liquids away from the console. Spills can cause short circuits on the exposed keypad and internal PCB tracks.
- Always place the device on a stable, flat desktop surface. Do not drop the console.
- Do not attempt to open the enclosure while the device is plugged in.

3. What's in the Box / Device Layout

- 1× Synex Digital Communication Visualization Console (DCVC) Unit
- 1× High-Stability AC-to-DC 12V Power Adapter
- 1× Official Warranty Card
- 1× DCVC Hardware User Manual



The console has the following main areas:

- 1) Left display (LCD1) — the main workflow screen: menus, tables, and the transmitting side of the simulation.
- 2) Right display (LCD2) — explanations, worked examples, algorithm traces, and the receiving side of the simulation.
- 3) Navigation buttons — five buttons used to move through every screen.
- 4) Keypad — used to type your message and enter numbers.
- 5) Noise dial — a knob that sets how noisy the simulated channel is.
- 6) LED Matrix — show the source coding.
- 7) TX LED bar — show the packet as it is transmitted.
- 8) RX LED bar — show the packet as it is received.
- 9) Packet transmission indicator
- 10) Packet receiving indicator
- 11) ACK transmission indicator
- 12) ACK receiving indicator

4. Controls

4.1 Navigation buttons

Button	What it does
OK	Confirms your current selection and moves forward.
Right →	Advances to the next major step (also used to page through result screens).
Left ←	Goes back to the previous screen.
Down ↓	Moves the selection to the next item in a list.
Up ↑	Moves the selection to the previous item in a list.

Tip: buttons respond the moment you press them; there's no need to hold them down.

4.2 Keypad

The keypad is used differently depending on which screen you're on:

- Message screen: press letters A–E to build your message; press “<” to erase the last letter.
- Probability entry / packet size screens: press number keys to type a value; “<” erases the last digit.
- CRC divisor screen: only 0, 1, and “<” are used, since a CRC divisor is a string of binary digits.

Other keys are simply ignored on screens where they don't apply, so there's no way to enter something invalid.

4.3 Noise dial

This dial sets the noise level of the channel. Turning it while a transmission is running changes the noise applied to the next packet, so you can see in real time how a noisier channel affects delivery.

4.4 LED indicators

The left LED bar shows the packet currently being transmitted (TX); the right bar shows the packets as received (RX). A separate indicator lights up while a packet is actively being sent or received, and another two LEDs light up to show an acknowledgement was sent and received, if you've turned ACKs on. The LED matrix display the source coding; turned on LEDs indicate 1 and blinking LEDs indicate 0.

5. Step-by-Step: Running a Transmission

This section walks through a complete run from power-on to a decoded message.

Step 1 — Enter your message

- 1. Power on the console. The message entry screen appears automatically.
- 2. Use the keypad to type a message using only the letters A–E.
- 3. Press “<” any time to correct a mistake.
- 4. Press OK to confirm your message.

Step 2 — Choose source coding

Source coding decides how your letters are turned into bits.

Option	What happens
Fixed Length	Every letter gets a ready-made 3-bit code. No further setup needed.
Huffman	You'll enter the probabilities of each letter (A–E), as percentages that add up to 100. The console then builds the Huffman code.
Shannon–Fano	You'll enter the probabilities of each letter (A–E), as percentages that add up to 100. The console then builds the Shannon-Fano code.

For Huffman or Shannon–Fano, after entering probabilities you'll choose Direct (jump straight to the final code table) or Step by Step (watch the code being built one decision at a time on the right-hand display).

Press Right to continue once you're viewing your final code table.

Step 3 — Choose channel coding

Channel coding adds extra bits so errors can be detected later.

Option	Notes
No Coding	No protection added; anything that arrives is accepted as-is.
Single-Bit Parity (Odd / Even)	Adds one extra bit per packet.
2D Parity	Arranges data in a small grid with row and column checks.
CRC-4	You'll enter a 5-bit divisor; a 4-bit checksum is appended.
CRC-8	You'll enter a 9-bit divisor; an 8-bit checksum is appended.

Select a method with Up/Down and confirm with OK.

Enter a packet size when prompted (the allowed range is shown on screen).

If you chose a CRC method, enter the divisor bits when prompted.
Review the resulting packet table, then press Right to continue.

Step 4 — Choose a modulation scheme

Pick BPSK, QPSK, ASK, PSK, or FSK. This changes only the on-screen waveform or constellation shown during transmission — it's a visualization, not a change to how errors are simulated.

Step 5 — Choose acknowledgements (ACK)

Choice	Behavior
Yes	A packet that fails its check is automatically retried, up to 20 extra attempts, until it passes or attempts run out.
No	Each packet is sent exactly once. If it fails its check, it's dropped.

Step 6 — Watch the transmission

Press OK to start. The console sends each packet in turn: it shows the transmitted bits, applies simulated noise based on the dial, shows what was actually received, checks it, and displays whether it was accepted, rejected, or retried. This runs automatically, no input is needed until it finishes.

Step 7 — Review the results

After the run completes, press Right to step through:

- Error-check review — the final pass/fail result for every packet.
- Padding review — how extra padding bits were removed before decoding.
- Decode review — the reconstructed message, letter by letter.

Because parity and CRC checks can't catch every possible error, the decoded message can occasionally differ from what you typed — that's expected, and part of what this console is designed to demonstrate.

Step 8 — Go again

The final screen lets you jump back to message entry, source coding, channel coding, or modulation without losing your other settings, so you can quickly compare, for example, the same message under two different noise levels or coding schemes.

7. Good to Know

- The alphabet is limited to letters A–E by design — this keeps the coding examples simple and easy to follow.
- All five letter probabilities must add up to exactly 100% before you can continue past that screen.
- Very long messages, combined with a small packet size, are limited by the console's internal capacity (up to 40 packets of up to 20 data bits each).
- A transmission run is a single continuous sequence — once started, let it finish before navigating elsewhere.

8. Maintenance

Cleaning the Enclosure: Wipe the chassis with a soft, dry microfiber cloth. Do not use chemical solvents or abrasive sprays, especially near the glass screens.

Powering Down: Unplug the 12V power adapter from the wall outlet if the console will not be used for an extended period.

9. Troubleshooting

Symptom	Likely cause / fix
A keypad press does nothing	That key isn't valid on the current screen (e.g. letters on a numeric-entry screen). Check the on-screen prompt.
Can't get past the probability screen	Your five percentages don't add up to exactly 100. Adjust and try again.
CRC divisor rejected	The divisor must be the correct length and must start and end with a 1 bit.
Decoded message doesn't match what you typed	Expected at high noise levels — not every error is caught by the selected channel coding. Try a stronger method (e.g. CRC-8) or lower the noise dial.
Up/Down doesn't scroll a review screen	Some review screens do not yet support scrolling; use Right to move to the next screen instead.

10. Warranty & Support

Legal Exclusions: Synex Engineering is not responsible for hardware damage caused by incorrect power supplies, liquid exposure, or unauthorized modifications. Furthermore, we are not liable for academic laboratory failures or grading issues related to the use of this device.

Limited Warranty Information: The DCVC unit comes with a 2-year limited warranty from the purchase date. This warranty covers internal manufacturing defects and component failures that occur under normal laboratory use.

The warranty will be immediately voided if:

- The enclosure is opened or modified.
- The device suffers physical impact damage (e.g., dropping).
- A third-party or incorrect power adapter is used.